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Declaration

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Abstract

Chapter 1

Gravitational Wave Interferometers

1.1 Introduction

The aim of this chapter is to outline some things about our universe we would like to know, how a gravitation wave allows this to be known, how an interferometer can be used to measure gravitational waves, the fundamental noise sources these detectors face. It aims to relate simple general-relativity based arguments to propose a gravitational wave detector. These noise sources can be reduced by making changes to the readout of the LIGO detector and by building new interferometers that use new technology. In this thesis, power of the laser and BHD is explored. Third generation detectors may operate at 2um, and so 2um detectors were characterised.

1.2 Astrophysical Origins of Gravitational Waves

BBH, BNS To date, gravitational waves emanating BBH (e.g. [2, 3]), BNS [4], and, possibly, from a binary consisting of a black hole and a neutron star (NSBH)

[5, 6] have been detected. The waveform of gravitational radiation emitted from these systems is known as a chirp signal. These signals increase in frequency as the two objects get closer to merging. Due to the geometry of these systems, the frequency of the gravitational waves are in the audioband with the cut-off frequency, determined by the radius of the objects that will merge, between 100 Hz-10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away from Earth, is $h \sim 10^{-21} - 10^{-23}$.

GR, what emits gravitational waves. Einstein's general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo gravitational wave detectors [16]. Gravitational waves are emitted from mass quadrupole systems (see e.g. [1])¹. As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses involved.

other sources of gravitational waves Other sources of gravitational waves include pulsars [7, 8] and magnetars [9]. Asymmetries in these systems will cause them to emit a gravitational wave with a constant frequency. Gravitational waves will be emitted by supernovae [10]. There will be a stochastic background of gravitational waves [11], similar to the cosmic microwave background radiation, as well as a background level of gravitational waves due to the merging of compact binaries [12]. Dark matter [13, 14] and cosmic strings [15] may give rise to gravitational waves.

¹Due to the conservation of mass/energy, there is no mass monopole radiation. Due to the conservation of momentum and angular momentum, there is no dipole radiation. The next term that can generate radiation is the quadrupole moment $h_{ij} = \frac{1}{R} \left(\frac{2G}{c^4} \ddot{I}_{ij} \right)$.

Observation Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 04/30/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	date?	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 1.1: table of observation runs

However, the gravitational waves from these systems are yet to be directly detected.

The need for increasing detections... To get better statistical models of BBH and BNS populations, multiple detections are required. To detect continuous wave signals, the signal to noise ratio improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, the gravitational wave detectors are operated for year long time-scales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades. The current observation runs are summed up in table 1.1.

1.3 Measuring Gravitational Waves with an Interferometer

Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and cross (e.g. [1]). The effect these waves have, in the local Lorentz frame, on test

masses on Earth is to move them by a small amount ².

To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. These are referred to as *end test masses* (ETMs). A sketch of a gravitational wave interacting with a Michelson interferometer is shown in Figure 1.1. For test masses L away from the origin of the local Lorentz frame, the distance they move ΔL due to an incoming plus polarised gravitational wave is

$$\Delta L = Lh. \quad (1.1)$$

The interferometer is sensitive to changes in the phase between the two arms corresponding of the order of one wavelength,

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} Lh, \quad (1.2)$$

allowing for small distances when laser light is used. The choice of $\lambda_{\text{EM}} = 1064 \text{ nm}$ as the wavelength which the detector operates with is determined by the low loss and low thermal noise coatings [19] and test masses that can be fabricated for 1064 nm light.

²In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses. This is due to the strong equivalence principle: all physical laws that are expressible as a tensor equation in special relativity have the exact same form in a local Lorentz frame of a curved space time **Reword?**. In the local Lorentz frame, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{2\pi L}{\lambda_{\text{EM}}}^2 h$. The Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = \frac{1}{2} hL$. Maxwell's equations, due to the strong equivalence principle, are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{2\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{2\pi L}{\lambda_{\text{EM}}}$ is sufficiently small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses. Alternatively, in the traceless-transverse gauge, the position of the test masses is fixed and the gravitational wave changes the wavelength of the light.

Gravitational wave detectors are operated on the *dark fringe* meaning that there is a low amount of power at the antisymmetric port for the signal light to beat with and the variation in the intensity of the light depends linearly on the gravitational wave strain. To do this, a small static differential offset, ϕ_0 , is introduced in the arm length. At the beam splitter, the electric field, Ψ_{AS} , emerging from the antisymmetric port due to the interference of the beams, with differential phase $\phi = \phi_0 + \Delta\phi$, returning from the end test masses, Ψ_y and Ψ_x , is

$$\Psi_{AS} = \frac{\Psi_y}{\sqrt{2}} - \frac{\Psi_x}{\sqrt{2}}, \quad (1.3)$$

$$= \sqrt{\frac{I_0}{2}} \left(\frac{e^{i(\phi_0 + \Delta\phi)}}{\sqrt{2}} - \frac{e^{-i(\phi_0 + \Delta\phi)}}{\sqrt{2}} \right). \quad (1.4)$$

The intensity of the beam at the anti-symmetric port is

$$I_{AS} = \Psi_{AS}^* \Psi_{AS}, \quad (1.5)$$

$$\simeq I_0 \phi_0^2 + 2I_0 \phi_0 \Delta\phi, \quad (1.6)$$

$$= I_0 \phi_0^2 + 2I_0 \phi_0 \frac{4\pi}{\lambda_{EM}} Lh, \quad (1.7)$$

In the absence of a gravitational wave signal, the DC light level at the antisymmetric port is related to the static offset by $\phi_0 = \sqrt{\frac{I_{AS}}{I_0}}$. As we are interested in a fluctuating signal, the DC offset from equation 1.7 can be removed, allowing for the signal at the antisymmetric port to be written as

$$I_{AS} = 2\sqrt{I_0 I_{AS}} \frac{4\pi}{\lambda_{EM}} Lh \quad (1.8)$$



1.3.1 Motivation for using a Fabry Perot Cavity in the Arms of a Michelson Interferometer

³Note that if the arm where to be made longer, the sensitivity of the detector to a signal at 100 Hz would decrease as the phase difference accumulated in the arms would no longer result in maximum constructive interference at the antisymmetric port.

The sites that were available to construct the current generation of gravitational wave detectors allow for 4 km arms. To reach the optimal arm length on a 4 km site, the number of times that the light bounces off the test mass, B , needs to be increased. Numerically, B should be ~ 250 . This means that equation 1.2 becomes

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}}BLh. \quad (1.9)$$

Fabry-Perot cavities are used⁴ to increase the number of bounces of the light from the test masses. The cavity is formed by placing an *initial test mass* (ITM) near the beam splitter. When a photon enters a cavity, it remains in the cavity for an average number of bounces that is determined by the reflectivity of the mirrors. When a Fabry-Perot cavity goes through a resonance, the phase of the reflected light rapidly changes in response to a change in arm length. This is illustrated in Figure 1.2. The interferometer measures $\Delta\phi_{\text{EM}} = \frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}} \Delta\phi_{\text{ETM}}$, so when a Fabry Perot cavity is near or on resonance, as the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$ is much greater for a Fabry Perot than it is for a mirror, the signal is enhanced. The exact way this happens is dependent of the frequency of the gravitational wave and the reflectivity of the mirrors, with the arm cavity cavity as a low pass filter (e.g see [17, Chapter 3] or [18, Appendix D]).

⁴Herriott delay lines were considered as a means of achieving more bounces from the test masses, however due to difficulty in manufacturing suitable mirrors which do not scatter light, thus generating noise, delay line interferometers fell out of favour. In principle, both delay lines and Fabry Perot cavities would perform equally well due to the Mizuno limit (see Section ??).

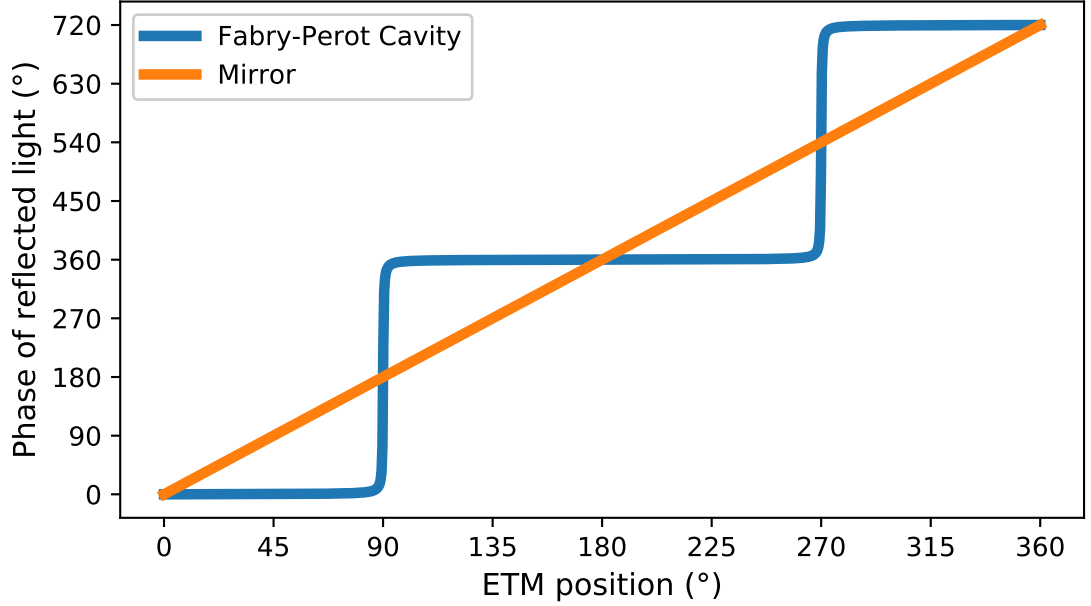


Figure 1.2: Comparison between the phase of the light reflected from a mirror versus the phase of light reflected from a cavity. The blue line shows the phase of the light that is reflected by a Fabry Perot cavity. Initially, the cavity is on anti-resonance so perfect destructive interference occurs as the round-trip phase is 180° . The end mirror (ETM) is moved 90° , resulting in an extra 180° round-trip phase, and so the cavity resonates. The phase of the light reflected from the cavity rapidly changes as it moves through the resonance. The ETM is moved further, the cavity stops being resonant and the phase of the light is not changed as its reflected by the cavity. The orange line shows the phase of the light when a mirror is used instead of a cavity. The interferometers sensitivity is proportional to the slop $\frac{d\phi_{EM}}{d\phi_{ETM}}$, and so by inserting another mirror in the arm near the beamsplitter, a Fabry-Perot cavity is formed and the sensitivity of the interferometer is increased.

1.4 Sources of noise in Gravitational wave Interferometers.

As the gravitational wave strain causes a minuscule change in the phase accumulated by light in the arm cavities, the challenge of detecting gravitational waves lies in suppressing the noise below this signal. From equation 1.9, for a gravitational wave of strain $h_{\text{rms}} = 10^{-23}$ interacting with LIGO the change in phase shift that the light experiences is $\sim 10^{-11}$ rad.

1.4.1 Shot Noise

The shot noise associated with detecting a coherent state, such as that created by the laser, can limit the sensitivity of the detector. To reduce the level of shot noise, the energy, E , of the laser needs to be high. The required energy to measure the phase shift that the light experiences due to the gravitational wave strain is determined by the time-energy uncertainty principle $\Delta\phi = \omega_{EM}\Delta T$ [20]. The measurement time, T , required corresponds to half the period of the gravitational wave $T = \frac{1}{2f_{\text{GW}}}$. The power of the laser is $I_0 = \hbar\omega_{EM}N$, where N is the number of photons detected in the time interval T , and so the energy used to measure the time is $E = I_0T = I_0\frac{\hbar\omega_{EM}N}{2f}$. As the laser is in a coherent state, the uncertainty in the number of photons detected in time T is $\Delta N = \sqrt{N}$. The uncertainty principle may now be written as

$$N \geq \frac{1}{4(\Delta\phi)^2}. \quad (1.10)$$

The amplitude spectrum of the intensity measured by the photodiode over a time corresponding to a bandwidth of Δf at the antisymmetric port will depend the

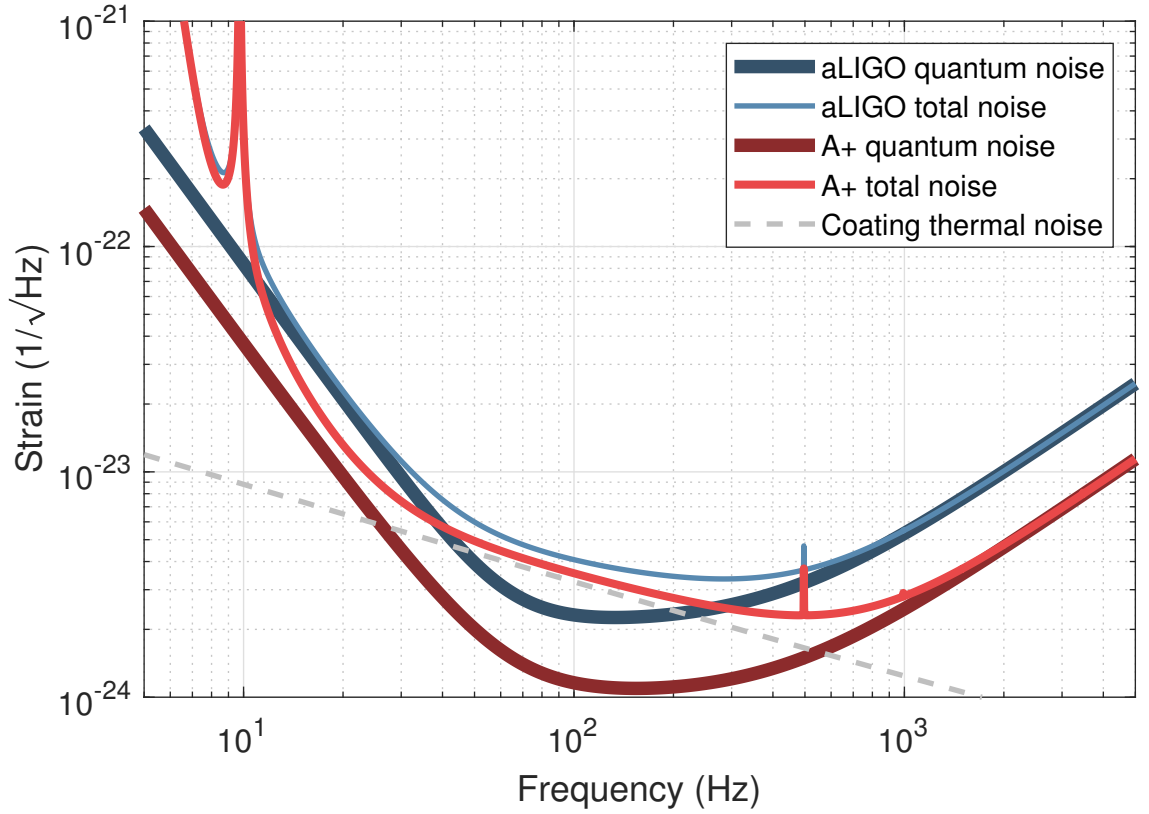


Figure 1.3: The Strain Sensitivity of the aLIGO and A+ detectors. The noise due to radiation pressure and photon shot noise is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise; with the other being the noise due to Brownian motion of the mirror's coating. Below 10 Hz, classical sources of noise, such as seismic motion, limit the sensitivity of both interferometers... figure out what is important to say here!

amount of light at the dark port,

$$I_{\text{ASrms}} = \sqrt{2\bar{I}_{\text{AS}}\hbar\omega_{EM}\Delta f}. \quad (1.11)$$

The conversion from light power at the antisymmetric port to gravitational wave strain can be inferred from Equation 1.8

$$\frac{h}{I_{\text{AS}}} = \frac{1}{2\sqrt{I_0\bar{I}_{\text{AS}}}} \frac{\lambda_{\text{EM}}}{4\pi LB} \quad (1.12)$$

and so the amplitude spectrum of the strain corresponding to the the shot noise of the light at the AS port is **simplify**

$$I_{\text{ASrms}} = \sqrt{\frac{\hbar\omega_{EM}}{2I_0}} \Delta f \frac{\lambda_{\text{EM}}}{4\pi LB}. \quad (1.13)$$

Equation 1.13 assumed the frequency response to a gravitational wave is flat, however the arm cavities have a low pass filtering effect on the signal, and so the shot noise will rise as f above the characteristic frequency of the arm cavity (as $B \rightarrow B(f) = B \frac{f_c}{f-f_c}$). This result is more rigourously found in [21].

To beat the photon shot noise by a factor of 10, the required number of photons to measure a gravitational wave with a strain amplitude of $h_{\text{rms}} = 10^{-23}$ is $\sim 10^{22}$. For a gravitational wave with a frequency of 100 Hz, the laser power required entering the arm cavity would need to be $\sim 1\text{MW}$. No such laser exists.

1.4.2 Power Recycling

To increase the laser power that is injected into the beam splitter, a power recycling cavity can be used [?]. Most of the light from the laser that is incident on the beam splitter gets reflected, effectively making the interferometer a mirror. Drever [?] proposed that a mirror, called the *power recycling mirror* (PRM), could be placed in the between the laser and the beam splitter and this would form a cavity. By tuning the position of the PRM such that the the light reflected from the interferometer and the light from the laser add in phase, the power would be ‘recycled’ and the total power impinging on the beam splitter would be increased. Current laser sources that can be operated for year-long time scales are capable of producing of the order 100W, and so the power recycling factor needs to be $\sim 10^4$. This technique has been successfully implemented by many large scale interferometers. [?]

1.4.3 Mode cleaning cavity to Remove Junk Light

mode cleaners remove junk light.

1.4.4 Signal recycling to shape quantum noise

Equation 1.13 shows that as the frequency of the the gravitational wave increases, the noise of the detection will increase. This detection bandwidth can be altered by introducing another mirror: the signal recycling mirror (SRM). This mirror is placed between the photodetector and the beamsplitter, and depending on its tuning, the bandwidth of the detector will change. More precisely, the SRM will form a coupled cavity with the arm cavities, but changing the tuning of the SRM, the finesse of the arm cavity as seen by signal carrying light will. There are two common modes of

operation: resonant sideband extraction (RSE) and signal recycling. In RSE mode, the signal recycling mirror is tuned so that the SRC is anti-resonant, thus broadening the sensitivity bucket at the expense of losing peak sensitivity. In signal recycling mode, the SRM is tuned so that SRC is resonant, deepening the sensitivity bucket at the expense of bandwidth. This comes from the conservation of energy. From an energy argument, the sensitivity of a detector to gravitational waves is determined by the detectors bandwidth and the energy within this bandwidth. This is known as the Mizuno limit [17].

1.4.5 Squeezed Light and the Standard Quantum Limit

squeezed light To reduce the shot noise of the limit of the detector, squeezed light can be used. The amplitude and phase of a light field have noise. In a coherent state, the quantum mechanical uncertainty in the phase quadrature and amplitude quadrature of the light is equal (**units**). The gravitational wave signal we are trying to measure is in the phase quadrature of the light, so it would be advantageous for, by the uncertainty principle, move noise to the undetected amplitude quadrature from the detected phase quadrature. This is an example of a squeezed state. In order to produce a squeezed state, a two photon process is required. This is realised with an optical parametric amplifier. This squeezed vacuum light is injected into the interferometer from the AS port so that the observed noise within the interferometer is squeezed. This technique has been implemented at LLO and GEO600 (citations).

radiation pressure noise So far, we have not considered the effect of light interacting with the test masses. In a quantum mechanical measurement, there is sensing noise (i.e. the shot noise), the zero point fluctuations of the probe (i.e. the test

masses acting quantum mechanically) and the noise due to the back action created by doing the measurement. In a Michelson interferometer, as a position measurement is being made, the light imparts a random amount of momentum into the test mass. This force noise becomes a displacement noise via a linear coupling, giving rise to radiation pressure noise. As this is a force noise and a position measurement is being made, its influence falls as $1/f^2$. At low frequencies radiation pressure noise dominates whereas at high frequency shot noise dominates. The relation between these two noises is called the standard quantum limit. Physically, this limit is set by the laser power. Increasing the laser power decreases shot noise but increases radiation pressure noise and vice versa. If the light is squeezed, then the radiation pressure noise will increase due to this coupling to the amplitude quadrature of light.

1.4.5.1 Light Quadratures and Balanced Homodyne readout

1.4.6 Techniques to beat Quantum Limits

Frequency dependent squeezing To avoid the increase in radiation pressure noise that is experienced when squeezed light is used, a frequency-dependent squeezer can be used. The light fields squeezing angle can be rotated as a function of frequency. This is done by using a detuned filter cavity. The light injected into it is rotated depending on its frequency. At low frequencies, the amplitude quadrature can be squeezed and the radiation pressure noise can be decreased. At high frequency, the phase quadrature can be squeezed and the shot noise can be decreased. This will be part of the A+ detector.

beating the bandwidth limit To beat this bandwidth limit, energy has to be injected into the system. White light cavities as a means of increasing the detector bandwidth have been studied (cite Haixing), although these solutions have yet to be demonstrated in the lab. Alternatively, PT-symmetry schemes may be used to beat the bandwidth limit, again these have yet to be demonstrated in the lab.

quantum non demolition There are several sub-SQL schemes. To avoid radiation pressure noise, a momentum measurement of the test masses could be made. As a speed measurement is being made, the displacement noise would rise as $1/f$ instead of $1/f^2$. A random impartation of ‘position’ would not affect future measurements. Such measurements are known as quantum non-demolition. **variational readout**

1.4.7 The Effect Losses have on Measuring Quantum States of Light

1.4.7.1 Photodiode Quantum Efficiency

1.4.7.2 Other losses

1.4.8 Other noise sources

There are other noise sources to contend with in an gravitational wave interferometer. The most notable of these is the noise associated with the Brownian motion of particles in the mirror coatings. Current detectors are designed around the expected noise level that can be achieved with these coatings. This will be one of the limiting sources of noise in the A+ detector. Other noise sources include seismic

noise from the motion of the Earth, Newtonian noise from pressure waves in earth crust, residual gas pressure noise that alters the refractive index of the arms. **Make list longer.**

1.5 The future of gravitational wave detectors

what limits current detectors (coating thermal noise arguments)

cyro As the limiting sources of of current gravitational wave detectors is the coating thermal noise, research into new mirror coatings is underway. As this is a thermal noise, the next generation of detectors may be operated at cryogenic temperatures. These coatings will not be compatible with the 1 um laser, so these detectors will operate with either 1.5 um or 2 um laser. Fused silica is opaque at these frequencies and does not improve when cooled, so silicon test masses are being developed. These new detectors will have much longer arm lengths than the current generation, and will be an order of 100 times more sensitive than A+. These detectors will be able to probe deeper into the universe than any other type of detector.

1.6 conclusion

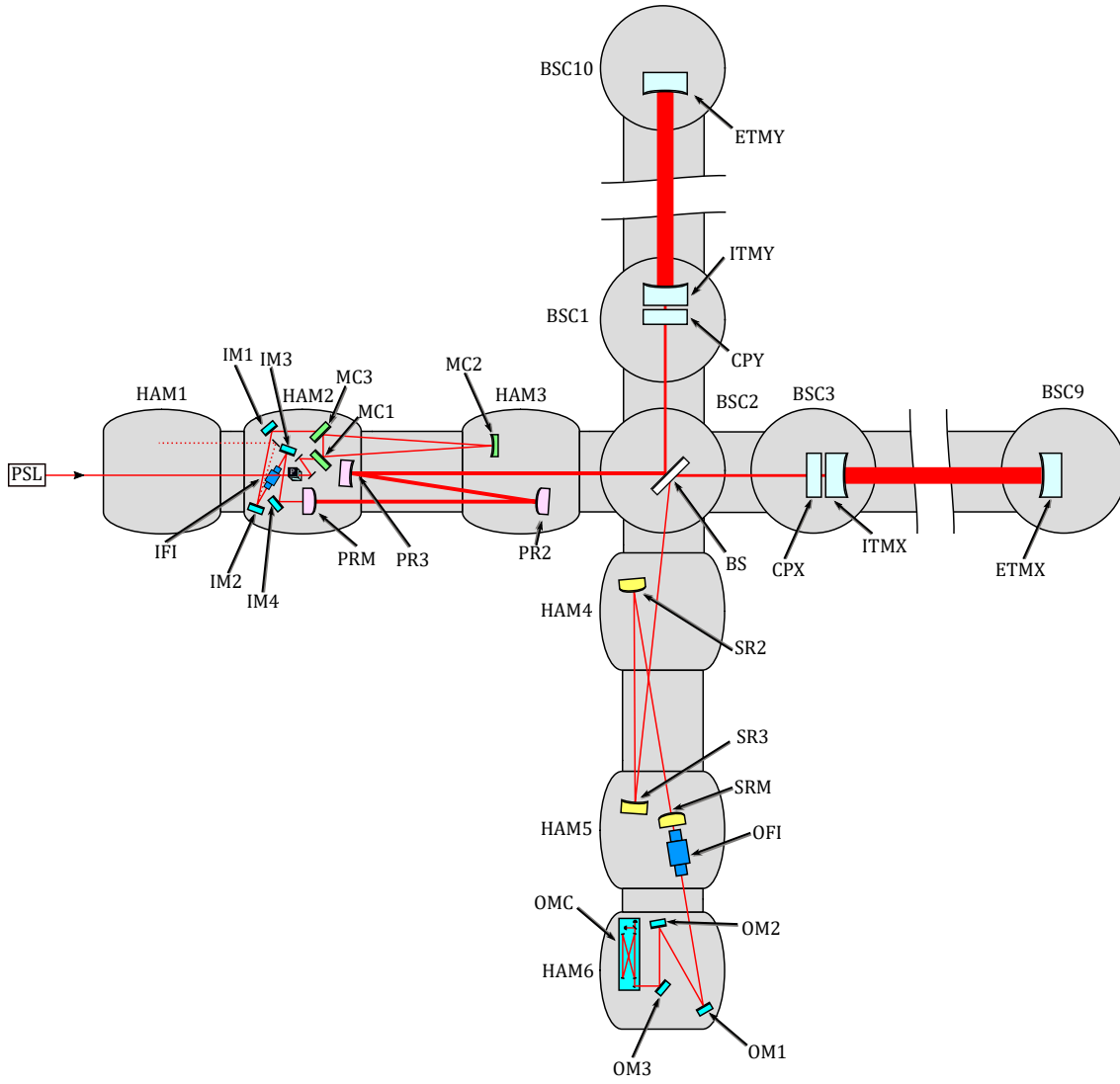


Figure 1.4

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